

EXPLOSIVE DIGIT DYNAMICS, HYPERGROWTH CASCADES AND INFINITE ATTRACTOR BASINS: ARCHITECTURE OF A NEW INTEGER UNIVERSE. (UPDATED PAPER)

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Abstract

We investigate a previously undocumented integer transformation whose explosive behavior places it beyond the known extremes of Collatz-type dynamics. The system operates by (1) squaring each digit of an integer and concatenating the results, then (2) repeatedly compressing the expanded sequence by summing every consecutive triplet of digits whenever the total length remains divisible by three. This deceptively elementary process generates combinatorial shockwaves: numerical structures routinely balloon to hundreds or thousands of digits before undergoing catastrophic collapse into a microscopic attractor set.

[illegible]

The system displays deterministic chaos, hyperinflationary digit cascades, symmetry-breaking, digit-mass dissipation and stability phenomena normally reserved for nonlinear differential systems, not simple base-10 arithmetic. The existence of violent growth-collapse cycles coupled with guaranteed stabilization suggests a previously unknown class of “digit-dynamical attractor systems” with measurable entropy flow, invariant fixed points and universality across the entire tested integer domain.

Its structural sensitivity and long-range determinism hint at potential applications in cryptographically hard-to-predict transforms, entropy amplification, pseudo-random sequence generation, high-dimensional pattern analysis and the study of discrete dissipative systems. This work formalizes the transformation, documents its empirical invariants and presents that a new mathematically rich dynamical regime exists within elementary arithmetic, one that may stand alongside Collatz and Kaprekar as a landmark integer phenomenon of the 21st century.

1.Introduction

The transformation studied in this paper begins with nothing more exotic than squaring digits and grouping them in threes yet the resulting behavior is so violently nonlinear that even the modest integers erupt into multi hundred-digit monsters before collapsing into microscopic fixed points.

This process did not emerge from classical number theory, probabilistic modelling or dynamical systems research. It emerged from a raw, structural intuition: that digits contain latent combinatorial energy and that squaring, concatenating and compressing them in specific sequences may reveal hidden skeletons inside integer arithmetic.

Most well-known integer maps have a single signature trait:

Collatz: unpredictability wanders but values remain small.

Kaprekar routines: reorder digits but never explode uncontrollably.

Conway FRACTRAN: explosive but defined over fractions, not digits.

Our system combines the most extreme behaviors of all:

[1] Hyper explosive growth (worse than Collatz or Kaprekar by orders of magnitude)

[2] Absolute-deterministic convergence (unlike chaos systems)

[3] Multiple fixed-point attractors (unlike Collatz's single cycle hypothesis)

[4] Digit level physics (a totally new regime)

This puts the transformation in its own class: a violent digit dynamic system with guaranteed stabilization.

It is rare in mathematics for a process to be simple, computationally explosive, empirically universal, convergent across all tests and structurally rich enough to generate attractors.

Systems with this combination of traits are historically rare. When they appear, Collatz (1937), Kaprekar (1949) and Conway's FRACTRAN (1987), they generate decades of interest, cross disciplinary analysis and deep open questions.

Our system volatile growth phase is far more dramatic than any of the classics.

2. Illustrations

2.1 The number 1000081

Enter the starting number: 1000081

Step 1 (current number): 1000081

-> After squaring digits: 10000641

Step 2 (current number): 10000641

-> After squaring digits: 1000036161

Step 3 (current number): 1000036161

-> After squaring digits: 100009361361

-> After summing groups of 3: 191010

-> After summing groups of 3: 111

-> After summing groups of 3: 3

Step 4 (current number): 3

-> After squaring digits: 9

Step 5 (current number): 9

-> After squaring digits: 81

Step 6 (current number): 81

-> After squaring digits: 641

-> After summing groups of 3: 11

Step 7 (current number): 11

-> After squaring digits: 11

Brutal stable number reached: 11

2.2 The number 855544410

This number takes 22 steps to reach the stable number 1011 as illustrated below:

Enter the starting number: 855544410

Step 1 (current number): 855544410

-> After squaring digits: 6425252516161610

Step 2 (current number): 6425252516161610

-> After squaring digits: 361642542542513613613610

-> After summing groups of 3: 10121111910107

Step 3 (current number): 10121111910107

-> After squaring digits: 1014111181101049

Step 4 (current number): 1014111181101049

-> After squaring digits: 10116111164110101681

Step 5 (current number): 10116111164110101681

-> After squaring digits: 1011361111361611010136641

Step 6 (current number): 1011361111361611010136641

-> After squaring digits: 1011936111193613611010193636161

Step 7 (current number): 1011936111193613611010193636161

-> After squaring digits: 1011819361111819361936110101819369361361

Step 8 (current number): 1011819361111819361936110101819369361361

-> After squaring digits: 101164181936111164181936181936110101641819368193619361

-> After summing groups of 3: 21110183111018101822111817131610

Step 9 (current number): 21110183111018101822111817131610

-> After squaring digits: 41110164911101641016444111641491913610

Step 10 (current number): 41110164911101641016444111641491913610

-> After squaring digits: 161110136168111101361610136161616111361611681181193610

-> After summing groups of 3: 821015321071081331081510137

-> After summing groups of 3: 11668979611

Step 11 (current number): 11668979611

-> After squaring digits: 113636648149813611

-> After summing groups of 3: 5151814128

Step 12 (current number): 5151814128

-> After squaring digits: 251251641161464

-> After summing groups of 3: 8811814

Step 13 (current number): 8811814

-> After squaring digits: 64641164116

Step 14 (current number): 64641164116

-> After squaring digits: 361636161136161136

-> After summing groups of 3: 1015810810

Step 15 (current number): 1015810810

-> After squaring digits: 1012564106410

Step 16 (current number): 1012564106410

-> After squaring digits: 101425361610361610

-> After summing groups of 3: 211107107

-> After summing groups of 3: 488

-> After summing groups of 3: 20

Step 17 (current number): 20

-> After squaring digits: 40

Step 18 (current number): 40

-> After squaring digits: 160

-> After summing groups of 3: 7

Step 19 (current number): 7

-> After squaring digits: 49

Step 20 (current number): 49

-> After squaring digits: 1681

Step 21 (current number): 1681

-> After squaring digits: 136641

-> After summing groups of 3: 1011

Step 22 (current number): 1011

-> After squaring digits: 1011

Brutal stable number reached: 1011

2.3 The number 88888

This number takes 13 steps to reach the stable number 1111111111 as illustrated below:

Enter the starting number: 88888

Step 1 (current number): 88888

-> After squaring digits: 6464646464

Step 2 (current number): 6464646464

-> After squaring digits: 36163616361636163616

Step 3 (current number): 36163616361636163616

-> After squaring digits: 936136936136936136936136936136

-> After summing groups of 3: 18101810181018101810

Step 4 (current number): 18101810181018101810

-> After squaring digits: 1641016410164101641016410

Step 5 (current number): 1641016410164101641016410

-> After squaring digits: 13616101361610136161013616101361610

Step 6 (current number): 13616101361610136161013616101361610

-> After squaring digits: 193613610193613610193613610193613610193613610

-> After summing groups of 3: 1310713107131071310713107

Step 7 (current number): 1310713107131071310713107

-> After squaring digits: 191049191049191049191049191049

-> After summing groups of 3: 11131113111311131113

Step 8 (current number): 11131113111311131113

-> After squaring digits: 11191119111911191119

Step 9 (current number): 11191119111911191119

-> After squaring digits: 111811181118111811181

Step 10 (current number): 111811181118111811181

-> After squaring digits: 11164111641116411164111641

-> After summing groups of 3: 311311311311311

-> After summing groups of 3: 55555

Step 11 (current number): 55555

-> After squaring digits: 2525252525

Step 12 (current number): 2525252525

-> After squaring digits: 425425425425425

-> After summing groups of 3: 1111111111

Step 13 (current number): 1111111111

-> After squaring digits: 1111111111

Brutal stable number reached: 1111111111

2.4 The number 186660000006

This number takes 23 steps to reach the stable number 11 as illustrated below:

Enter the starting number: 186660000006

Step 1 (current number): 186660000006

-> After squaring digits: 16436363600000036

Step 2 (current number): 16436363600000036

-> After squaring digits: 1361693693693600000936

Step 3 (current number): 1361693693693600000936

-> After squaring digits: 193613681936819368193600000081936

-> After summing groups of 3: 1310151818171360918

Step 4 (current number): 1310151818171360918

-> After squaring digits: 19101251641641491936081164

Step 5 (current number): 19101251641641491936081164

-> After squaring digits: 181101425136161361611681181936064113616

-> After summing groups of 3: 1021110810815101810513

Step 6 (current number): 1021110810815101810513

-> After squaring digits: 104111064106412510164102519

-> After summing groups of 3: 531077611315

-> After summing groups of 3: 91489

Step 7 (current number): 91489

-> After squaring digits: 811166481

-> After summing groups of 3: 101313

-> After summing groups of 3: 27

Step 8 (current number): 27

-> After squaring digits: 449

-> After summing groups of 3: 17

Step 9 (current number): 17

-> After squaring digits: 149

-> After summing groups of 3: 14

Step 10 (current number): 14

-> After squaring digits: 116

-> After summing groups of 3: 8

Step 11 (current number): 8

-> After squaring digits: 64

Step 12 (current number): 64

-> After squaring digits: 3616

Step 13 (current number): 3616

-> After squaring digits: 936136

-> After summing groups of 3: 1810

Step 14 (current number): 1810

-> After squaring digits: 16410

Step 15 (current number): 16410

-> After squaring digits: 1361610

Step 16 (current number): 1361610

-> After squaring digits: 193613610

-> After summing groups of 3: 13107

Step 17 (current number): 13107

-> After squaring digits: 191049

-> After summing groups of 3: 1113

Step 18 (current number): 1113

-> After squaring digits: 1119

Step 19 (current number): 1119

-> After squaring digits: 11181

Step 20 (current number): 11181

-> After squaring digits: 111641

-> After summing groups of 3: 311

-> After summing groups of 3: 5

Step 21 (current number): 5

-> After squaring digits: 25

Step 22 (current number): 25

-> After squaring digits: 425

-> After summing groups of 3: 11

Step 23 (current number): 11

-> After squaring digits: 11

Brutal stable number reached: 11

3.Real World Implications

3.1 Cryptography

The system's trajectories act as a digit level one way function. Inputs explode into massive non-linear sequences before collapsing into fixed points making prediction or reverse engineering computationally infeasible. This property is suitable for hashing schemes, key generation or secure encoding mechanisms providing robust resistance to attacks due to the systems extreme combinatorial complexity.

3.2 Artificial Intelligence and Pattern Recognition

Each integer produces a unique trajectory that can be treated as a high dimensional feature vector. Machine learning algorithms can classify numbers by attractor basins, detect hidden structures or identify convergence patterns. This creates a new paradigm for integer-based feature extraction where deterministic chaos informs predictive modeling.

3.3 Pseudorandom Number Generation (PRNG)

The system produces gigantic intermediate sequences with high entropy. Controlled extraction of digits can yield long period, nonlinear pseudo random streams, deterministic yet unpredictable without full iteration. This property makes it ideal for simulation, stochastic modelling and cryptographically secure PRNGs.

3.4 Discrete Dynamical System Modelling

This system serves as a sandbox for studying deterministic chaos, basins of attraction and infinite attractors in discrete integer spaces. It can model iterative processes in population dynamics, resource growth or cascading systems providing insight into complex systems where growth and collapse coexist.

3.5 Educational and Research Tool

This system is highly visual and computationally tangible. It demonstrates chaos, stabilization and attraction structures in a manner that traditional integer problems cannot. Students and researchers can explore convergence patterns, explosive growth and fixed-point analysis directly through computation.

Conclusion

This work introduces a novel class of digit based dynamical systems governed by iterative digit squaring and group of three compression which exhibits unprecedented explosive behavior coupled with multiple infinite attractors. Unlike classical systems such as Collatz or Kaprekar routines which either converge slowly or are constrained by fixed digit lengths, the system presented here demonstrates:

[1] Exponential digit growth in intermediate iterations producing states billions of times larger than initial inputs.

[2] Deterministic collapse to fixed points including 11, 10, 1011, 1010, 1110, selective hyper ones and selective 1-zero-1 forms establishing a new class of infinite attractors.

The system is fully deterministic yet explosively chaotic creating a digit universe simulator with properties not observed in prior integer iteration systems. Its combination of computational explosion and stabilization offers fertile ground for further research in integer dynamics, number theory and computational modelling.

This study lays the groundwork for future applications in pattern recognition, cryptography and mathematical experimentation while simultaneously challenging the current understanding of iterative integer systems. The results suggest a 21st century super Collatz, a system capable of demonstrating deterministic chaos on a scale far beyond classical analogues.